

Exercises in Marine Ecological Genetics

04. Genome sequencing and assembly

- Short- and long-read sequencing
- Raw sequencing data
- Genome assembly
- Command-line basics

Martin Helmkamp

<https://github.com/mhelmkampf/meg26>



Image by Jake Wilton / Ocean Image Bank

Course outline

Preliminary, may be subject to change

Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	<i>F</i> -statistics and population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II (optional)	11_edna.R

F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to F -statistics.

- How would the number of heterozygotes be affected by self-fertilization?
- What would you expect for the inbreeding coefficient F_{IS} in this case?
- Use the *H. puella* dataset ("caribbean") to check your prediction



Photograph by Luiz Rocha

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F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to F -statistics.

```
### Calculate global F-statistics
genepop::Fst("data/msats/puella_caribbean.gen",
            outputFile = "work/Fst_caribbean.txt")

Fwc(is) All: 0.0023    # see bottom of output file
```

- Use the *H. puella* dataset ("caribbean") to check your prediction
- F_{IS} is very low, **no evidence for self-fertilization**

Today's objectives

By the end of this session, you will be able to:

- Compare **major genome sequencing technologies** and their key trade-offs
- Inspect and process raw sequencing data in **Fastq format**
- Generate a **draft genome assembly** and assess assembly quality using contiguity and completeness metrics
- Use **basic bash commands** to navigate files and execute simple assembly steps

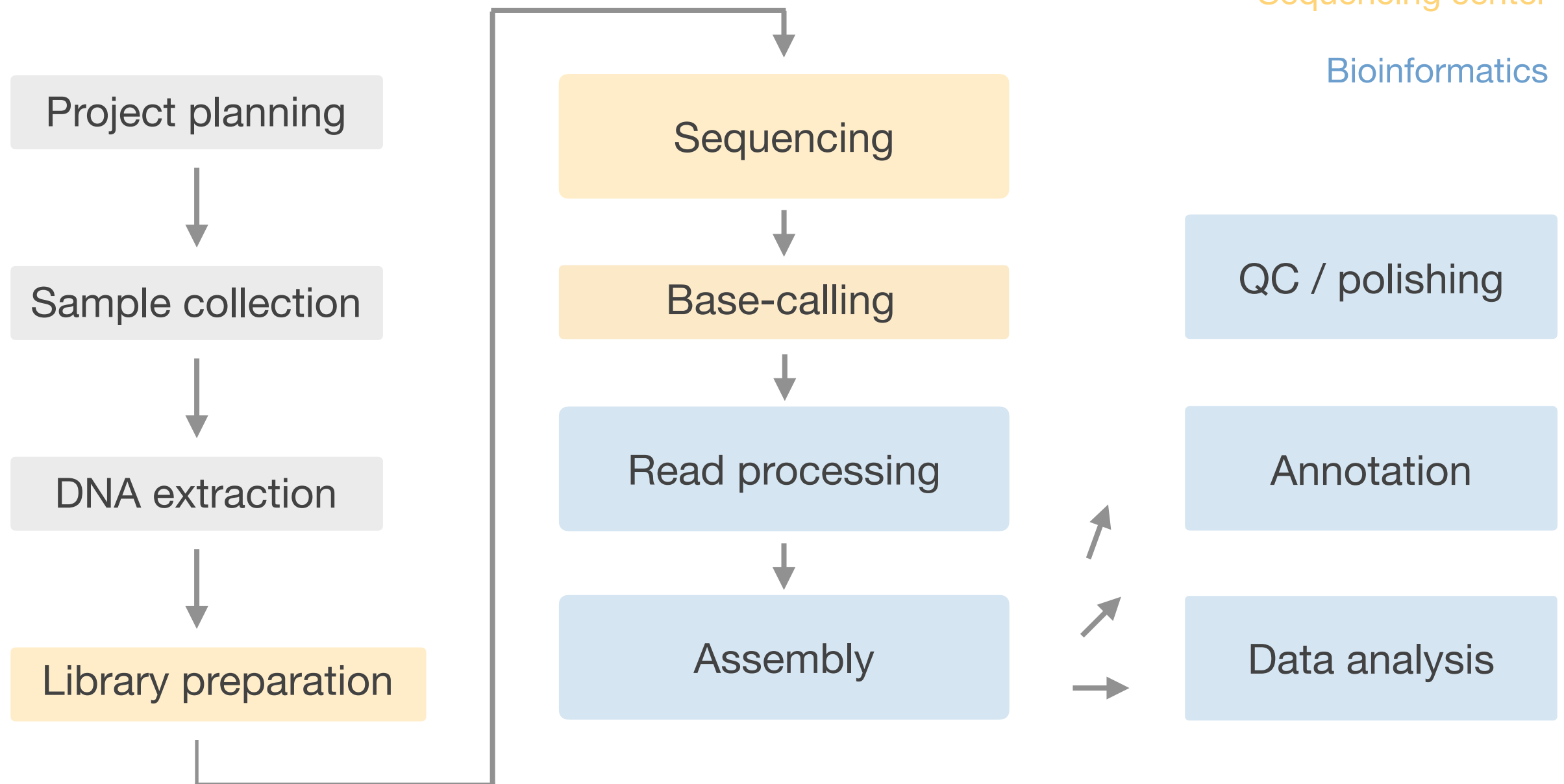
De novo genome sequencing workflow

Legend

Preparation

Sequencing center

Bioinformatics



Exercise 1

```
head -n 4 HypPue1_illumina_raw_F.fastq # display first 4 lines of file
```

```
@HWI-ST1293:199:HA9JHADXX:1:1101:1044:1603 1:N:0:CGATGT  
NCCCTGTTAAAGGATCATCTCTGACCTATCATTGTGGTGTAAATCACATTTAACACAATCACGATGTGCTTTACCTGCAGC  
ATCTTTACAGCAGGGCTGGGAGATATGACC AAAACAGTTATGATAATATGTTTTATTTCTATTGAAAATCA  
+  
#1=DDFFHHHHHHJJJJJJJJJJJJJJJJJJJJJJGHIIJJJJHJJJJJJJJJJJJJJJJJJJJHHHHH  
FFFFFFEEEEEDDDDDDDDD@DDDEDEDDBDDDDDDDCDCEDDEEDEEEDEECDDDEDEDEEDDDDDDDDC
```

1. @ followed by sequence id and optional info (e.g. instrument/run id, barcode)
2. DNA sequence
3. +, sometimes followed by sequence id
4. base quality score (same length as sequence)

Base quality

Phred **quality score**:

$$Q = -10 \log_{10} P$$

Common benchmark:

% bases with $Q \geq 30$

Quality score	<i>P</i> incorrect base call	Base call accuracy
10	1 in 10	90 %
20	1 in 100	99 %
30	1 in 1000	99,9 %
40	1 in 10000	99,99 %

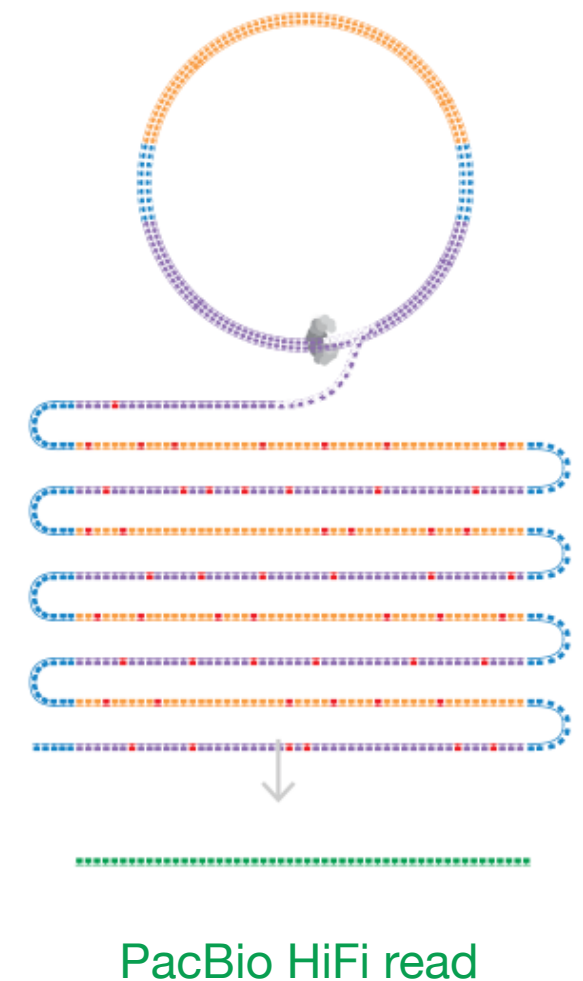
FASTQ encoding (Illumina 1.8+):

ASCII Symbol:	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJ
Quality Score:	0.2.....10.....20.....30.....41

Genome assembly

Exercise 2

- Reconstructing long, continuous sequence from millions of overlapping **reads**
- Reads can be very short (e.g. Illumina) or long (e.g. PacBio, ONT)
- Segments of assembled sequence are called **contigs**, which may be combined into scaffolds
- Scaffolds or PacBio contigs can be up to chromosome-length



Sequencing technologies compared

Exercise 2

Technology	Read length	Accuracy	Gb per run	Human genome	Devices
Illumina	2 × 150 bp	> 99.9 %	1000–10000	300–700 €	NextSeq 2000, NovaSeq X
PacBio	10–20 kb	> 99.8 %	80–300	1200–2500 € ↓	Sequel II, Revio
ONT	10–100 kb	97–99 % ↑	50–200	900–2200 € ↓	MinION, PromethION

As of 2026; typical values in real-world academic settings; ↓ ↑ trends

bp = base pairs, kb = kilo bases (1000 bp), Mb = Mega bases (million bp), Gb = Giga bases (billion bp)

Human genome: cost for 30 × coverage, including library prep (Sonnet 4.6 estimate)

Sequencing technologies compared

PacBio

Illumina

- High yield, low cost, most accurate
- Short reads limit contiguity
- ▶ Whole genome resequencing (population genomics), eDNA / metabarcoding

- Long reads, accurate
- Expensive, requires high-quality DNA
- ▶ High-quality reference genome assemblies, structural variation, phasing

ONT

- Portable, very long reads possible
- Less accurate
- ▶ eDNA / metabarcoding, projects in low-resource settings



Photo: Achim Meyer

Assessing assembly quality

Exercise 2

- Sequencing depth / coverage
- Assembly metrics: size distribution of contigs / scaffolds
- Average base accuracy (Q score)
- Percentage of assembly assigned to chromosomes
- Gene completeness
- Phasing information

Challenges

- Contamination
- Misassembled regions
- Presence of false duplications

Sequencing depth / coverage

Exercise 2

- Average number of reads representing each position in the genome
- coverage or depth = $\text{read count} \times \text{average read length} / \text{genome size}$
- high coverage facilitates assembly, detection of sequencing errors
- Typical coverage: 50–100 × (or more) for *de novo* genome sequencing

10–30 × for re-sequencing

Genome: CGTAATGGCATATCGCCTAGATTTCGAAACG

Read 1: TAATGGCATATCGCCTAGAT

Read 2: CATATCGCCTAGATTTCGAAA

Read 3: TATCGCCTAGATTTCGAAACG

Depth: 00111111223333333333333322222211

- Total size (compare to expected genome size)
- Number of contigs / scaffolds
- Largest scaffold
- **N50**: contig / scaffold size where 50% of assembly is found on contigs / scaffolds of equal or larger size (measure for fragmentation / sequence contiguity)

Scaffolds: 530, 760, 1050, 610, 450, 800, 220, and 1200 kb

Reorder: 1200, 1050, 800, 760, 610, 530, 450, 220 kb

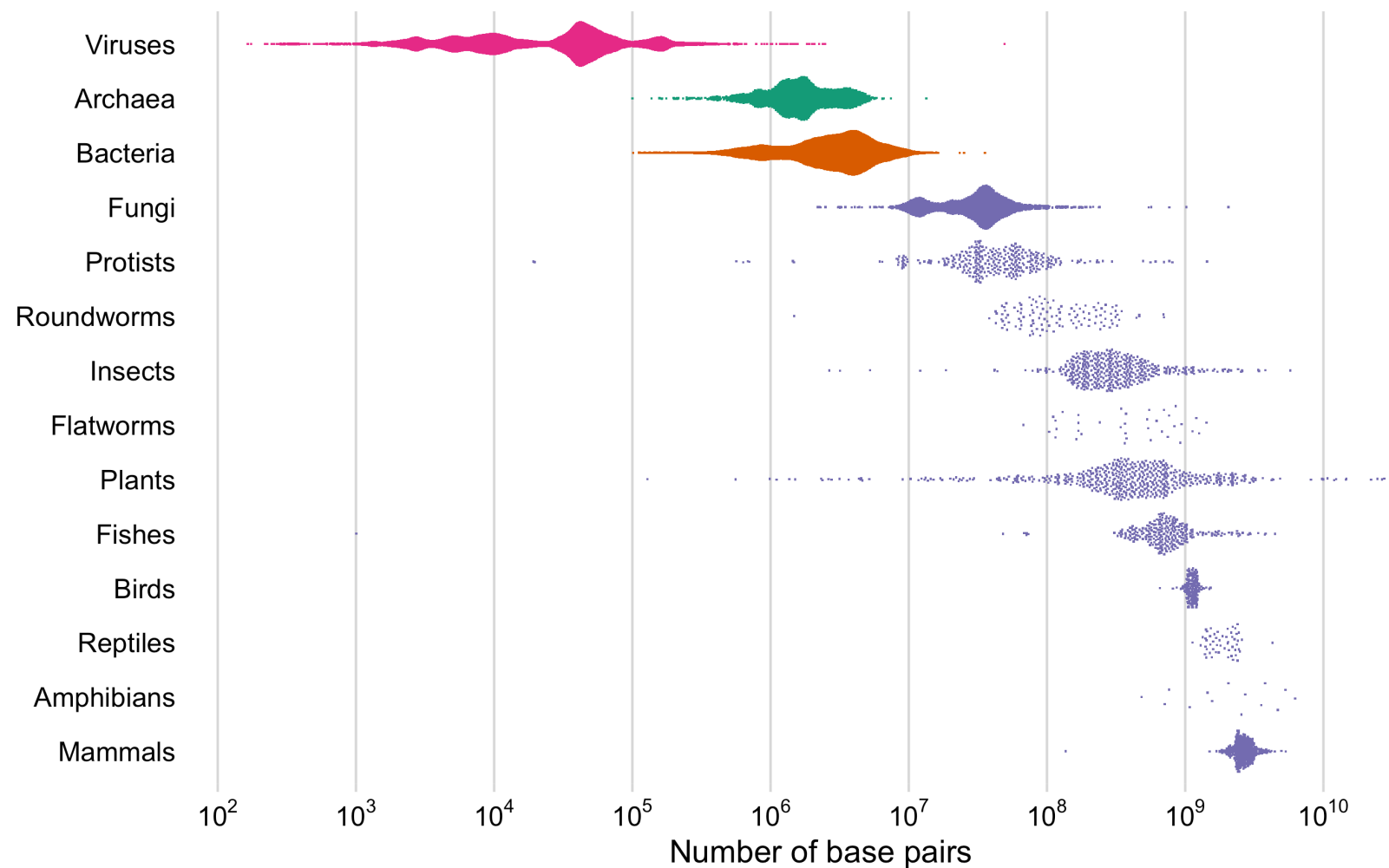
Sum/2: $5620/2 = 2810$

Add up until sum/2 is reached: $1200 + 1050 + 800 > 2810$

N50 = 800 kb

Genome size

Exercise 2



- Nine orders of magnitude
- Genome size correlated with repetitive DNA, which is hard to sequence and assemble

Based on 50,000 organisms with genome information in NCBI

See also www.genomesize.com

Tom E. White, Datavision 2020

The Blackchin guitarfish genome

Exercise 2/3

PacBio HiFi sequencing of *Glaucostegus cemiculus*

Assembly using hifiasm

Kirsti Burnett

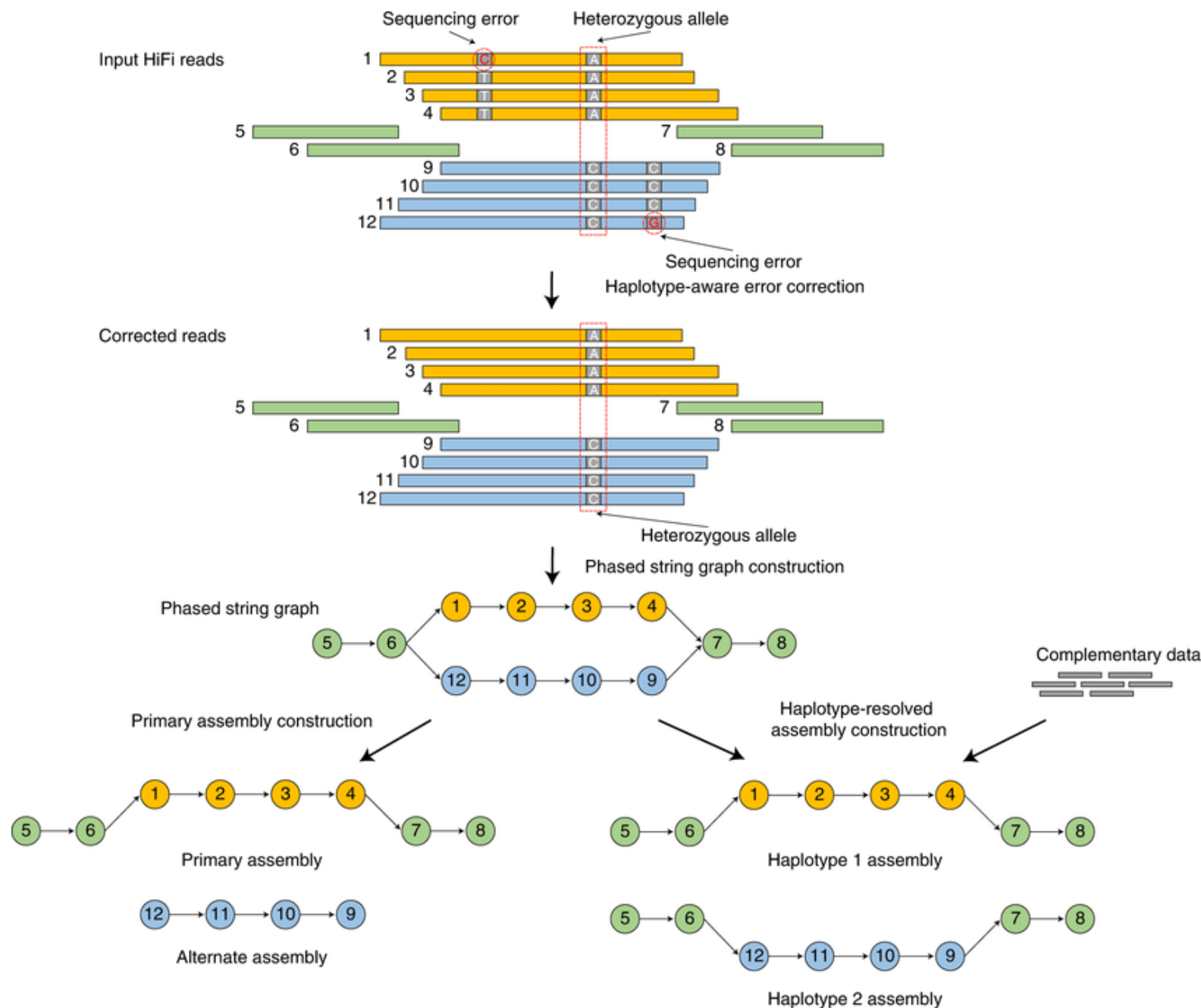


Genome assembly stats

Size	2.85 Gb
No. contigs	1511
N50	11.9 Mb
Coverage	18 ×

Primary vs. haplotype assemblies (hifiasm)

Exercise 3

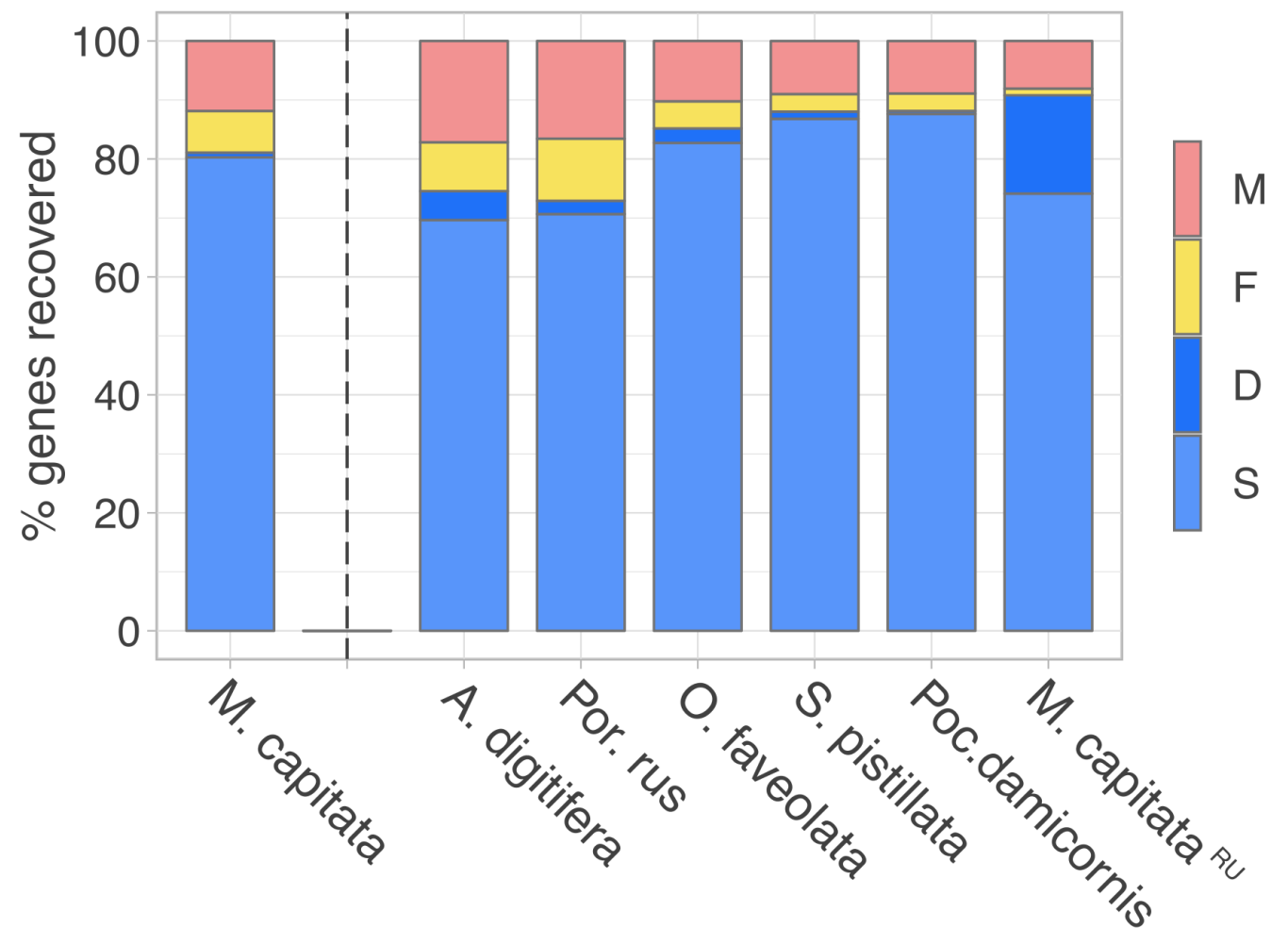


Cheng et al. 2021 *Nature Methods*

Gene completeness with BUSCO

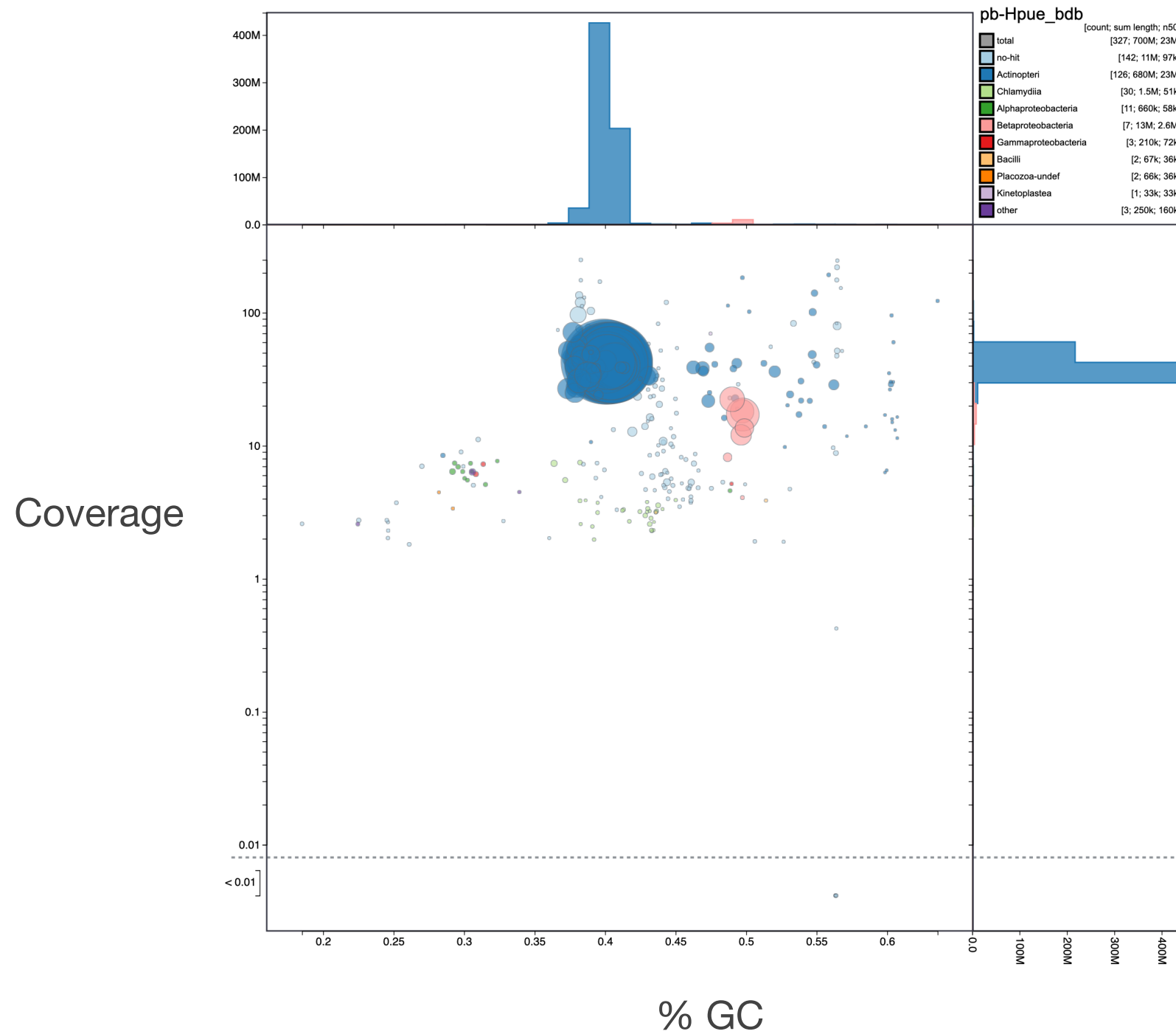
<https://busco.ezlab.org>

Quantifies assembly completeness based on presence of universal, highly conserved, single-copy genes (e.g. housekeeping genes)



Helmkamp et al. 2019 (Genome Biology and Evolution)

Contamination QC with BlobTools



Color:
Most similar
known taxon

HypPue2.1_pacbio_pctg.fas

Basic bash commands

Summary

<code>ls -l</code>	# List directory contents in long format
<code>cd <directory></code>	# Change into directory
<code>cd ..</code>	# Change into directory above / back
<code>mkdir <directory></code>	# Create directory
<code>mv <file/dir> <file/dir></code>	# Move or rename file or directory
<code>cat <file></code>	# Print file to screen
<code>head -n 100 <file></code>	# Print first 100 lines of file

<code>tab key</code>	# Autocomplete file or directory name
<code>up arrow key</code>	# Cycle through previous commands

Genome sequencing and assembly

Summary

```
grep -c '^+$' <raw_reads>.fastq      # Calculate number of reads in Fastq file  
  
assembly_stats <assembly_file>.fas    # Calculate assembly metrics (Python tool)  
  
hifiasm -o <output_dir> --primary <read_file>.fastq    # Assemble PacBio reads
```

- **Short- and long-read sequencing** technologies offer different but complementary strengths
- **Assembly quality** can be evaluated using metrics like coverage, contiguity (e.g. N50) and completeness (e.g. BUSCO)
- **Assemblies are drafts** – often fragmented and with errors – but serve as the foundation for further analyses
- Genome-scale data have **many applications**, including generating large SNP datasets for population genetic analyses, inferring population history, and detecting genes under selection